

Selective Shift: Towards Personalized Domain Adaptation in Multi-Agent Collaborative Perception

Hui Zhang
School of Computer Science
and Technology,
Beijing Jiaotong University,
Beijing, China
huizhang1@bjtu.edu.cn

Yidong Li *
School of Computer Science
and Technology,
Beijing Jiaotong University,
Beijing, China
yqli@bjtu.edu.cn

Yiteng Xu
School of Computer Science
and Technology,
Beijing Jiaotong University,
Beijing, China
24125346@bjtu.edu.cn

Tiago H. Falk
INRS-EMT,
University of Quebec,
Montreal, Canada
tiago.falk@inrs.ca

Yonglin Tian
Institute of Automation,
Chinese Academy of Sciences,
Beijing, China
tyldyx@mail.ustc.edu.cn

Fei-Yue Wang*
DeSci Center of Parallel Intelligence,
Obuda University,
Budapest, Hungary
Institute of Automation,
Chinese Academy of Sciences,
Beijing, China
feiyue.wang@ia.ac.cn

Abstract

Given the scarcity of real data and the time-intensive nature of labeling, current multi-agent perception models often rely on simulated sensor data for training and validation. However, perception performance deteriorates significantly due to domain gap between simulated and real data. Existing adaptation methods focus on domain-generalized feature extraction while neglecting multi-agent shift uncertainty and relational semantic loss. To address this issue, we propose a **Selective Shift Domain Adaptation** method in multi-agent collaborative perception, called SSDA. SSDA incorporates two essential components: the frequency-decoupled feature shift adjustment (FSA) and the entropy-driven staged adaptive alignment (SAA). To mitigate the relational semantic loss, the FSA is proposed to simplify the representation of correlation features and remove redundant information from the source domain, thereby mitigating interference for domain adversarial scenarios. To tackle the shift uncertainty, the SAA is designed to achieve adaptive alignment from global to local guided by information entropy, which dynamically adjusts weights for samples according to their level of uncertainty. The results demonstrate that the SSDA is significantly superior to the SOTA, achieving up to 7.35% improvements on AP@0.7.

CCS Concepts

• **Computing methodologies** → **Object recognition**.

Keywords

Collaborative Perception, 3D Object detection, Domain Adaptation

*Corresponding author



This work is licensed under a Creative Commons Attribution 4.0 International License.
MM '25, Dublin, Ireland

© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2035-2/2025/10
<https://doi.org/10.1145/3746027.3754723>

ACM Reference Format:

Hui Zhang, Yiteng Xu, Yonglin Tian, Yidong Li, Tiago H. Falk, and Fei-Yue Wang. 2025. Selective Shift: Towards Personalized Domain Adaptation in Multi-Agent Collaborative Perception. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3746027.3754723>

1 Introduction

In recent years, autonomous driving has emerged as a rapidly evolving field with the potential to transform transportation systems, enhancing both safety and efficiency on the roads. One of the key components in autonomous driving is perception, which serves as the backbone for situational awareness, providing crucial data about the vehicle's environment to support decision-making and control. As perception technologies continue to evolve, Vehicle-to-Everything (V2X) collaborative perception has garnered substantial attention in both academia and industry [24, 27, 31, 44, 47]. By enabling communication and cooperation among multiple vehicles and infrastructure elements, V2X collaboration can mitigate the limitations of individual sensors, such as constrained detection ranges and sensor blind spots. This cooperation significantly enhances the perception capabilities of autonomous vehicles, enabling more accurate and reliable environmental awareness.

While V2X collaborative perception offers considerable advantages, developing a high-accuracy V2X detection model for autonomous driving still demands extensive annotated real-world data, which is both costly and time-intensive to obtain. Synchronizing multiple sensors, setting up scenarios, and labeling the collected data require substantial manual effort. In contrast, creating collaborative perception data within a simulation environment proves far more economical and efficient. Large-scale simulated datasets can be generated autonomously on simulation platforms, such as CARLA [10], and PreScan [5]. However, collaborative perception involves multiple agents (such as autonomous vehicles, and infrastructures) working together to perceive their surroundings by

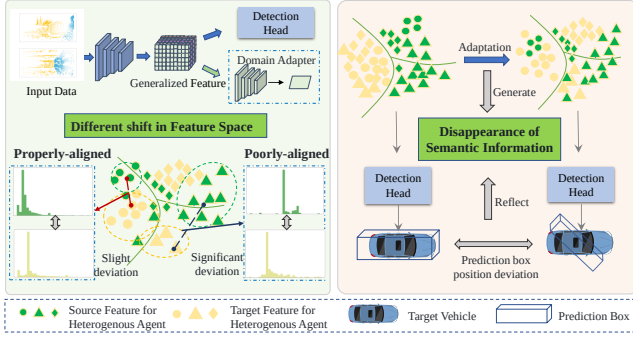


Figure 1: Key challenges in UDA for collaborative perception: shift uncertainty and relational semantic loss.

sharing and integrating data. Variations between domains, such as environmental conditions, sensor types, or data distributions, make direct application of models trained in one domain ineffective in another. Consequently, achieving effective transfer of models across domains remains a critical challenge.

Unsupervised domain adaptation (UDA) tackles the challenge of transferring knowledge across different environments or sensor setups where labeled data is limited or unavailable in the target domain. Existing methods [16, 18] attempt to directly use the adversarial learning strategy to mitigate the simulated-to-real gap and multi-agent gap. These methods concentrate on domain-generalized feature extraction but neglect the uncertainty of domain shift across different heterogeneous agents in collaborative perception. Although they enable direct alignment between real and virtual domains, the bias resulting from shift uncertainty causes the loss of crucial semantic information, thereby affecting the model’s perception performance. As illustrated in Figure 1, there is significant variance in the real-virtual gap across heterogeneous agents due to visual appearance, object diversity, and environmental dynamics. Some agents require slight deviation for alignment, while others may need significant deviation. This uncertainty in transfer leads to the bias of feature shift, and the bias in feature adaptation consequently results in the loss of key semantic information. The loss leads to errors in position coordinates and angles in the detection results.

Building on these insights, we introduce a selective shift method towards personalized domain adaptation in multi-agent collaborative perception, called SSDA. As shown in Figure 2, SSDA incorporates two essential components: the frequency-decoupled feature shift adjustment (FSA) and the entropy-driven staged adaptive alignment (SAA). The FSA module captures the frequency relationships and unravels the intricate relationships between features from different agents in the frequency domain. Frequency decoupling via FSA effectively simplifies the key information in the original input. Relevant key information is preserved while domain-specific interference from the source domain is removed. It consists of frequency characteristics decoupling, adaptive feature refinement, and regularization adjustment.

The SAA module progressively adjusts the alignment through two stages: the Global Alignment Stage (GAS) and the Local Alignment Stage (LAS). The GAS focuses on global feature alignment

from different domains, while the LAS focuses on aligning local features from multiple collaborative agents. Meanwhile, the entropy-guided attention map is introduced to measure the offset among the heterogeneous agents and allow the adaptive adjustment of adversarial loss weights based on whether each sample is properly-aligned or poorly-aligned.

Our contributions can be summarized as follows:

- We introduce SSDA, a selective shift domain adaptation method in multi-agent collaborative perception, which divides the domain adaptation into two problems: shift uncertainty and relational semantic loss.
- The FSA is proposed to precisely extract the lost correlation information during agent alignment. By acquiring frequency features with wavelet transform, it simplifies the representation of correlation features and removes redundant information from the source domain, thereby mitigating interference for domain adversarial scenarios.
- The SAA is developed to alleviate the shift uncertainty by information entropy-guided adaptive alignment from global to local. It allocates varying weights to samples based on their level of uncertainty.
- We perform comprehensive experiments from simulated to real-world datasets. The results demonstrate the effectiveness of our method and highlight the essential role of the proposed components.

2 Related Works

2.1 Collaborative Perception

Collaborative perception improves the performance of autonomous vehicles by enabling information sharing between multiple agents, allowing them to acquire a more complete understanding of their surroundings. [37]. Previous studies have made significant progress in both effective communication and adaptive fusion [14, 25, 26, 43, 51] under perfect conditions, as well as in mitigating real-world challenges such as communication redundancy and transmission delay [13, 15, 54]. In the face of data paucity and the uncontrollability of real-world environments, most research is validated on simulated datasets, such as OPV2V [49], V2X-Sim [19], and V2XSet [48]. However, virtual data fails to accurately represent the intricate interactions and noise found in real-world environments, leading to significant discrepancies when compared to real-world data. To deal with this issues, existing methods [16, 18] introduce a domain adaptation method for V2X collaborative detection, aiming to achieve feature alignment between simulated and real-world data. However, they do not consider the feature disparity bias among different agents and semantic information loss caused by feature shift [1, 2]. Unlike the aforementioned works, we focus on the differences in internal feature shifts and the decoupling of key features. A method based on information entropy is proposed for selective shift, achieving superior performance.

2.2 Domain Adaptation

UDA is designed to enhance model performance on unlabeled target domains by drawing on shared knowledge from labeled

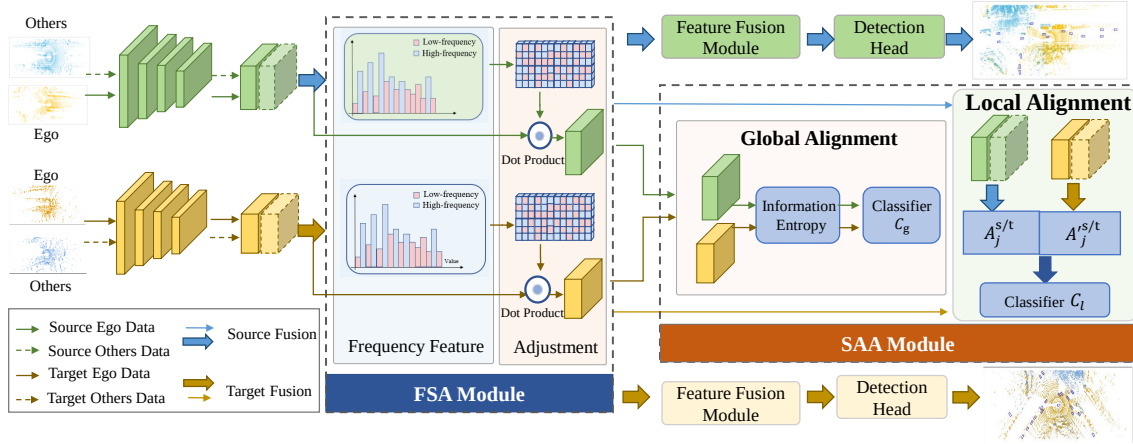


Figure 2: Framework of the proposed SSDA method. SSDA incorporates two essential components: the FSA and the SAA module. The FSA module captures the frequency relationships and unravels the intricate relationships between features from different agents in the frequency domain. The SAA module allocates varying weights to samples based on their level of uncertainty and alleviates the shift uncertainty by information entropy-guided adaptive alignment from global to local.

source domains [3, 4, 8, 9, 28, 35]. Built on the underlying principles, relative advancements can be broadly categorized into two approaches: adversarial learning [12, 23, 36, 40, 56] and metric learning [22, 39, 41, 50]. CMDA [6] introduces an unsupervised domain adaptation approach based on adversarial learning to mitigate domain discrepancies. CDASA [55] presents a selective dual-adversarial network capturing both domain-invariant and category-discriminative features. Based on metric learning, DADA [33] gives a domain adaptation technique to address the domain discrepancy between sample clusters and proxy sets. Lai *et al.* [17] explore an effective adaptation method for vision-language models that preserves existing knowledge while enhancing their ability to integrate new information. CUDA-X [52] establishes a unified optimization architecture for unsupervised end-to-end cross-domain generalization through joint enhancement of semantics, localization, and feature representations. CoPEFT [45] enables swift adaptation to novel scenarios via parameter-efficient hierarchical adaptation mechanisms. However, metric learning methods rely on learning task-specific distance metrics, making their performance highly sensitive to the given dataset and failed to handle the high sparsity and irregularity of point cloud data. Therefore, we propose an adversarial learning strategy that effectively extracts global and local information from point clouds for feature alignment.

3 Method

UDA involves transferring knowledge from a source domain, where labeled data is available, to a target domain that lacks labeled data. In the context of collaborative perception, this problem becomes more complex, since the agents may use heterogeneous sensors with different data types and placements. Each collaborative agent contributes diverse sensor data, such as LiDAR point clouds, along with its pose information in the global coordinate system. The goal of domain adaptation is to effectively leverage the labeled data from the source domain to improve the model’s performance on the target domain, where data distribution differs due to various factors like sensor noise, resolution, and environmental mismatches.

Overview: The proposed SSDA can be utilized in most typical collaborative 3D detectors that rely on intermediate fusion. Figure 2 shows the framework of SSDA. The collaborative detection module typically involves fusing intermediate-level feature representations from multiple sensors or agents before making the final classification scores and bounding box regression. The FSA module utilizes wavelet transform to extract the characteristics of different agents in the spatial frequency domain, thereby generating attention weights in the frequency domain and finely aligning the features to readjust the shift. It is applied before feature fusion, with the input consisting of the features from each agent. The SAA module performs adaptive alignment from both global features of different domains and local features of various collaborative agents. It is applied after the detection head, with the input consisting of the classification scores and features for each agent.

Notation: Let N_0^s denote the ego agent from the source domain and N_0^t denote that from the target domain. Accordingly, $\Omega(N_0^s)$ and $\Omega(N_0^t)$ respectively denote the other agents collaborating with the ego agent in the source and target domains. Let $\mathbb{X}_s = \{(x_{N_0}^s, x_{N_i}^s, y_i^s), N_i \in \Omega(N_0^s)\}$ denote the labeled data from the source domain, where $\{(x_{N_0}^s, x_{N_i}^s)\}$ is the input collaborative sample and y_i^s is the corresponding label. Similarly, let $\mathbb{X}_t = \{(x_{N_0}^t, x_{N_i}^t), N_i \in \Omega(N_0^t)\}$ represent the unlabeled data from the target domain, where $\{(x_{N_0}^t, x_{N_i}^t)\}$ is the input collaborative sample from the target domain and no labels are provided. The superscript ‘s’ and ‘t’ denote the source and target domains, respectively, while ‘s/t’ indicates that it could be either a source or target domain. The FSA is employed prior to feature fusion, where the input comprises the features from each agent $F_j^{s/t}, j \in N_0^{s/t} \cup \Omega(N_0^{s/t})$. The SSA is applied after the detection head, with the input including the classification scores $S_j^{s/t}, j \in N_0^{s/t} \cup \Omega(N_0^{s/t})$ and features from each agent $F_j^{s/t}, j \in N_0^{s/t} \cup \Omega(N_0^{s/t})$. The goal is to learn a model that can generalize well on both the source and target domains.

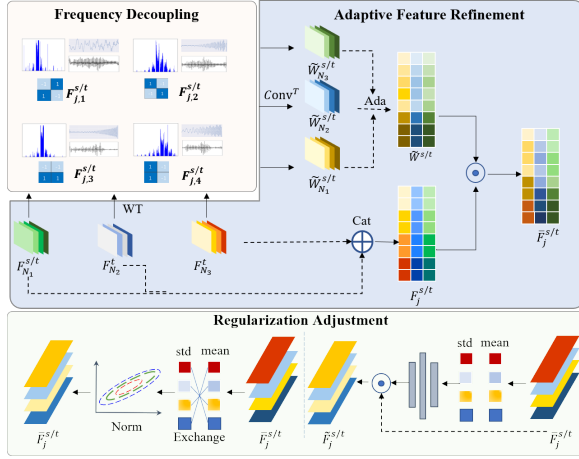


Figure 3: Architecture of the FSA module. The FSA can be partitioned into three components: frequency characteristics decoupling, adaptive feature refinement, and regularization adjustment.

3.1 Frequency-decoupled Feature Shift Adjustment

Current researches focus on aligning data across domains through domain adaptation such as metric learning [11, 30] and adversarial learning [16, 29], which effectively enhances the robustness of the model. However, when these methods are applied to collaborative perception tasks, the effects are insignificant. The main reason is that there are complex dependencies between the features of different agents, and directly transferring the agents' features to the ego feature may result in the loss of important semantic information. In domain adaptation studies, frequency-adaptive learning schemes that decompose visual features into low- and high-frequency subspaces have become mainstream [21, 38]. However, these methods often overemphasize low-frequency components while indiscriminately suppressing high-frequency details, leading to an irreversible loss of edge textures and anatomical structure semantics. This imbalance in feature representation results in models with strong spatial localization but compromised semantic classification performance. To address this issue, the FSA module proposed in this paper effectively integrates both high- and low-frequency information, filtering out domain-irrelevant details while preserving essential features to enhance discriminative power.

The FSA can capture the frequency relationships among different agents, as shown in Figure 3. By decoupling in the frequency domain, it becomes possible to more effectively disentangle the complex interdependencies between features from different agents, leading to a clearer and more independent feature representation. Diverging from conventional low-frequency domain-invariant feature extraction strategies, FSA leverages the intrinsic spectral correlations between low- and high-frequency signals to establish a spectral mapping framework, enabling precise reconstruction of high-frequency critical information. The proposed method capitalizes on the inherent low-frequency signal's dual role as both a carrier mechanism for high-frequency semantic information and a

stabilizing element through spectral coupling properties. By strategically preserving high-frequency salient features during transmission while maintaining low-frequency structural integrity, this optimized signal partitioning enables enhanced modality-specific feature extraction. Such separation not only mitigates cross-modal interference but also establishes a foundation for hierarchical feature alignment across heterogeneous agents, thereby significantly improving the coherence and precision of collaborative perception systems.

The FSA can be partitioned into three components: frequency characteristics decoupling, adaptive feature refinement, and regularization adjustment. The details are as follows:

Frequency Characteristics Decoupling. Given an agent feature ($F_j^{s/t}, j \in N_0^{s/t} \cup \Omega(N_0^{s/t})$) derived from the neural network, we exploit the sparsity properties of wavelet transforms to achieve two critical objectives: (1) reducing computational complexity by leveraging the sparse distribution of wavelet coefficients and (2) mitigating domain discrepancies through noise suppression across both low- and high-frequency components. This is effectively realized through the application of the Haar Wavelet Transform (HWT). Specifically, $F_j^{s/t}$ passes through four kinds of filters to separate low-frequency and high-frequency features, each of which applies a depth-wise convolution with a stride of 2 on two dimensions. High-frequency details are particularly critical in addressing point cloud detection in collaborative perception. Therefore, FSA utilizes three high-pass filter banks and one low-pass filter to efficiently extract essential information. Different filtered features are generated to reduce the feature redundancy, represented as:

$$F_{j,k}^{s/t} = \text{HWT}^k(F_j^{s/t}), \quad (1)$$

where $\text{HWT}^k, k \in \{1, 2, 3, 4\}$ denotes the transform with the k -th filter, highlighting the frequency-domain characteristics of each agent's features $F_j^{s/t}$. The reconstructed feature for each filter of the j -th agent is computed by the transposed convolution:

$$\hat{F}_{j,k}^{s/t} = \text{Conv}^T(F_{j,k}^{s/t}, w_{j,k}^k). \quad (2)$$

Adaptive Feature Refinement. Adaptive feature refinement integrates high-frequency and low-frequency information to generate weights for adjusting the shift of features adaptively. The reconstructed high-frequency features and low-frequency features are combined by:

$$\tilde{W}_j^{s/t} = \text{Cat}(\hat{F}_{j,k}^{s/t}, k \in \{1, 2, 3, 4\}). \quad (3)$$

After reconstructing the frequency features, the agents generate weight maps to foreground the semantic information from different spatial regions. Notably, an excessive adjustment focusing on the initial information can influence the feature shift results. Therefore, we simplify the weight map with an adaptive fusion strategy:

$$\tilde{W}^{s/t} = \text{Ada}(\tilde{W}_j^{s/t}, j \in N_0^{s/t} \cup \Omega(N_0^{s/t})), \quad (4)$$

where $\tilde{W}^{s/t}$ denotes the weight map employed to modify feature distribution, j denotes the j -th agent in the collaborative sample. $\text{Ada}(\cdot)$ denotes a dynamic feature-weighting mechanism, which adaptively integrates the feature weight maps via spatially localized

convolutions to generate a globally optimized weight map. The features of each agent are then refined by the weight map:

$$\tilde{F}_j^{s/t} = \tilde{W}^{s/t} \odot F_j^{s/t}, \quad (5)$$

where $\tilde{F}_j^{s/t}$ denotes the adjusted features, which is applied not just for 3D detection but also for domain adaptation. By this way, it can avoid the hindrance of effective feature transfer caused by the input of excessive source domain information that has not been transferred.

Regularization Adjustment. To address the issues of internal covariate shift and the statistical characteristic disparities between different domains, the FSA introduces statistical attribute obfuscation (SAO) and statistical feature weighting (SFW). SAO adopts a normalization approach that exchanges the statistical parameters between different features. Specifically, the mean and variance are calculated to capture the distributional disparities of features across different domains. After that, we exchange the statistical parameters between features to normalize them, which is denoted by:

$$\tilde{F}_j^{s/t} = \frac{\tilde{F}_j^{s/t} - \mu_{\pi(j)}}{\sigma_{\pi(j)}}, \quad (6)$$

where $\mu_{\pi(j)}$ and $\sigma_{\pi(j)}^2$ denote mean and variance of $\pi(j)$. $\pi(j)$ randomly swaps the feature indices of agent j with those of another agent.

In this way, the features abandon their inherent distributional properties of the original domain and are encouraged to adapt within the new domain. Therefore, the model concentrates more on the universal characteristics of the data and reduces the interference caused by domain discrepancy. Additionally, SFW is similar to adaptive feature refinement. We use the computed statistical parameters to weight the features from different domains, helping to minimize the loss of semantic details in the statistical features.

$$\tilde{F}_j^{(s/t)} = \omega_j \cdot \tilde{F}_j^{(s/t)}, \quad (7)$$

where $\tilde{F}_j^{(s/t)}$ denotes weighted feature, ω_j is generated through an affine transformation of the input feature $\tilde{F}_j^{(s/t)}$.

Through the three core components, the FSA module first performs frequency characteristics decoupling to separate domain-sensitive noise from semantic critical information. Then, it adaptively optimizes cross-agent feature distribution via adaptive feature refinement. Finally, it mitigates inter-domain statistical discrepancies via regularization adjustment, effectively achieving the dual goals of preserving high-frequency details while suppressing domain-specific noise.

3.2 Entropy-driven Staged Adaptive Alignment

We propose a SAA module that assesses sample information entropy across different domains and utilizes it for adaptive sample alignment progressively, as illustrated in Figure 4. The SAA module gradually adapts the alignment by integrating the GAS and the LAS, where the GAS aligns global features across different domains and the LAS aligns local features from various collaborative agents.

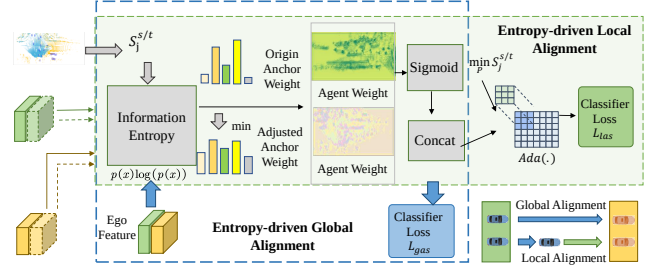


Figure 4: Architecture of the SAA module. It gradually adapts the alignment by integrating the GAS and the LAS. The GAS facilitates cross-domain alignment, guided by an entropy-based attention map of global domain features. The LAS achieves alignment across different agents, guided by an entropy-based attention map of local agent features.

Global Alignment Stage. The global image feature of ego agents from different domains ($F_{N_0}^s$ and $F_{N_0}^t$) are fed into the global-level domain classifier (C_g). The domain classifier consists of the gradient reverse layer and two 1×1 convolutional layers. The entropy-driven attention map is generated to guide the alignment from global domain level. According to Shannon entropy, higher entropy indicates greater difficulty in predicting the concealed information, signifying lower information purity. Consequently, the entropy of poorly-aligned samples is greater than that of properly-aligned ones. To prioritize the alignment of poorly-aligned samples while preventing the excessive alignment of properly-aligned ones, we propose the attention-guided adversarial learning strategy leveraging an entropy-aware attention map. The attention map is generated based on the entropy of the classification score outputs, defined as:

$$S_{N_0}^{s/t} = \Phi_c^{s/t}(F_{N_0}^{s/t}; \theta_c^{s/t}), \quad (8)$$

$$\tilde{A}_{N_0}^{s/t} = S_{N_0}^{s/t} * \log(S_{N_0}^{s/t}) - (1 - S_{N_0}^{s/t}) * \log(1 - S_{N_0}^{s/t}), \quad (9)$$

where Φ_c denotes the detection head of the collaborative detector. $S_{N_0}^{s/t} \in \mathbb{R}^{B \times P \times H \times W}$ denotes the classification scores in source or target domain. B is the number of ego agents and P is the number of proposals for each sample in the batch size. $A_{N_0}^{s/t} \in \mathbb{R}^{B \times P \times H \times W}$ denotes the global attention map in source or target domain. To estimate prediction confidence, we select the proposal with the lowest entropy at each location.

Local Alignment Stage. The LAS concentrates on distinguishing the fine-grained feature maps ($F_j^{s/t}$, $j \in N_0^{s/t} \cup \Omega(N_0^{s/t})$) among heterogeneous agents while disregarding the domain gap between different domains. The domain classifier (C_l) takes heterogeneous agent features as input, consisting of the gradient reverse layer and two 1×1 convolutional layers. The LAS module considers the alignment between heterogeneous agents from two perspectives: the information entropy perspective and the data volume perspective. Following by GAS, the information entropy is employed to assess the uncertainty of agents and adaptively adjust the adversarial loss weighting for different agents. Besides, some grids are nearly empty,

containing only a sparse number of points. As a result, these grids offer limited information about the sensors. Supervision on C_l for these grids may result in noisy gradients, which could exacerbate the instability of the optimization process. For the information entropy perspective, the entropy attention map $A_j^{s/t}$ is formulated as follows:

$$A_j^{s/t} = S_j^{s/t} * \log(S_j^{s/t}) - (1 - S_j^{s/t}) * \log(1 - S_j^{s/t}), \quad (10)$$

$$A_j^{s/t} = \min_P (A_j^{s/t} + \max(A_j^{s/t}))/2, \quad (11)$$

where $S_j^{s/t} \in \mathbb{R}^{P \times H \times W}$ denotes the classification score of j -th agent, and $\min_P$ denotes minimization over the parameter dimension P to refine the attention weights. For the data volume perspective, the volume attention map $A_j^{s/t}$ is obtained as:

$$A_j^{s/t} = \min_P C_j^{s/t}(F_j^{s/t}; \theta_c^{s/t}), \quad (12)$$

After that, the local-level attention map $\tilde{A}_j^{s/t}$ is generated by combining $A_j^{s/t}$ and $A_j^{s/t}$ adaptively, formulated as follows:

$$W_j^{s/t} = \text{Ada}(\text{Cat}(A_j^{s/t}, A_j^{s/t})), \quad (13)$$

$$\tilde{A}_j^{s/t} = W_j^{s/t} \times A_j^{s/t} + (1 - W_j^{s/t}) \times A_j^{s/t}, \quad (14)$$

where $\text{Cat}(\cdot, \cdot)$ represents the concatenation of input features along the channel dimension. $\text{Ada}(\cdot)$ is implemented by a 1×1 convolutional layer.

By integrating the GAS and LAS, SAA facilitates adaptive cross-domain feature optimization. GAS uses gradient reversal layers to enforce global feature alignment while dynamically adjusting adversarial loss weights according to classification confidence. LAS combines information entropy and data distribution density to manage differences in features across multiple agents. It achieves detailed feature alignment by generating localized attention maps. Through the global and local alignment stages, the method effectively mitigates cross-domain distribution variances and boosts collaborative perception stability.

3.3 Training Objectives

The overall training objective of SSDA consists of three components: the detection loss L_{det} , the GAS loss L_{gas} , and LAS Loss L_{las} .

Detection Loss: The prediction loss consists of two major components: a classification loss L_{cls} , which is a focal loss [34], and regression loss L_{reg} , which is a smooth-L1 loss [46]. The overall detection loss is denoted by:

$$\mathcal{L}_{det} = \mathcal{L}_{cls} + \alpha \mathcal{L}_{reg}. \quad (15)$$

Global Alignment Loss: We introduce an entropy-guided adversarial learning approach to align features between different domains. Given the ego features from the source and target domains (F_{ego}^s and F_{ego}^t), the global-level entropy-guided adversarial loss is defined as follows:

$$\mathcal{L}_{gas} = \sum_{B, H, W} [\tilde{A}_{N_0}^s \mathcal{L}_e(C_g(F_{N_0}^s), g^s) + \tilde{A}_{N_0}^t \mathcal{L}_e(C_g(F_{N_0}^t), g^t)], \quad (16)$$

where $\mathcal{L}_e$ denotes the cross entropy loss, B is the batch size. g^s and g^t denote the pixel-level domain label (i.e., 0 for source domain and 1 for target domain).

Local Alignment Loss: Given agent-level features $\tilde{F}_j^{s/t}$, the local-level attention map $\tilde{A}_j^{s/t}$, and the local-level domain classifier C_l , the local-level entropy-guided loss is defined as:

$$\mathcal{L}_{las} = \sum_{N^s, H, W} \tilde{A}_j^s \mathcal{L}_e(C_l(\tilde{F}_j^s), g^s) + \sum_{N^t, H, W} \tilde{A}_j^t \mathcal{L}_e(C_l(\tilde{F}_j^t), g^t), \quad (17)$$

where N^s and N^t denote the number of agents from source domain and target domain respectively. Therefore, the overall loss function of the proposed SSDA is defined as:

$$\mathcal{L}_{SSDA} = \mathcal{L}_{det} + \beta_1 \mathcal{L}_{gas} + \beta_2 \mathcal{L}_{las}, \quad (18)$$

where β_1 and β_2 denote the parameter factors corresponding to global alignment and local alignment.

Training Optimization: The training objective of the proposed SSDA is:

$$F^*, C_g^*, C_l^* = \underset{F}{\text{argmin}} \max_{C_g} \max_{C_l} \mathcal{L}_{SSDA}. \quad (19)$$

4 Experimental

4.1 Datasets and Metric

Datasets: To validate the effectiveness and generalizability of the SSDA, we evaluate the proposed SSDA on the task of collaborative detection using V2XSet [48] and DAIR-V2X-C [53].

V2XSet [48] is an open virtual dataset for V2X collaborative perception. The dataset covers V2X collaboration across different road types and is designed to evaluate the performance of collaborative 3D object detection.

DAIR-V2X-C [53] is the first large-scale 3D object detection dataset for vehicle-road collaboration research, with all data collected from real-world scenarios. Similar to the V2XSet, the collection equipment includes a single vehicle and a roadside infrastructure. Therefore, we primarily use V2XSet as the source domain and DAIR-V2X-C as the target domain to train a domain adaptation model that bridges the virtual and real worlds.

Evaluation Metric: In the evaluation, we test on the DAIR-V2X-C dataset and report the average precision (AP) at interaction ratio (IoU) thresholds of 0.3, 0.5, and 0.7.

4.2 Implementation Details

We implement the proposed SSDA using different detectors on the Pytorch platform [32], utilizing two NVIDIA RTX 4090 GPUs for training. The initial learning rate is 0.01, and it decays every 15 epochs by a factor of 0.1. The Adam optimizer is employed to optimize the model. β_1 is set to 0.5 and β_2 is set to 1. α is set to 2.5. For each iteration, we input two batches of data simultaneously, one from the source domain and the other from the target domain.

4.3 Quantitative Evaluation

Table 1 shows the results of the SSDA as compared with baselines in terms of AP. Compared to the state-of-the-art domain adaptation strategy DUSA [16], the proposed SSDA demonstrates outstanding

Methods	V2X-ViT [48]			F-Cooper [7]			DiscoNet[20]		
	AP@0.3	AP@0.5	AP@0.7	AP@0.3	AP@0.5	AP@0.7	AP@0.3	AP@0.5	AP@0.7
Oracle	62.69	59.02	45.61	61.26	57.19	45.04	62.87	58.53	46.69
No Adapt	34.73	31.23	14.94	35.12	30.27	13.46	35.11	31.07	16.93
Self-training	33.03	31.67	19.01	40.46	35.89	16.38	30.66	28.72	17.26
Discriminator	38.54	33.89	15.13	41.33	36.20	16.81	39.55	34.37	19.05
DUSA [16]	43.61	38.46	20.09	42.83	37.93	20.25	40.03	35.23	20.68
No Adapt w/ GAS	46.85(+3.24)	43.72(+5.26)	28.51(+8.42)	48.45(+5.62)	43.82(+5.89)	24.30(+4.05)	50.56(+10.53)	47.07(+11.84)	30.81(+10.13)
SSDA w/o FSA	50.91(+7.3)	46.42(+7.98)	24.84(+4.75)	49.64(+6.81)	45.70(+7.77)	28.00(+7.75)	49.70(+9.67)	45.86(+10.63)	30.34(+9.66)
SSDA	50.76(+7.15)	45.73(+7.27)	27.44(+7.35)	54.30(+11.47)	49.69(+11.76)	29.18(+8.93)	50.53(+10.50)	46.47(+11.24)	30.38(+9.70)

Table 1: Overall performance on different detectors compared with the state-of-the-art methods. The results are reported in AP@0.3/0.5/0.7. Bold type indicates the optimal metric, and the red text signifies the greatest enhancement. ‘+’ shows the increase in performance compared to the DUSA [16]. ‘w/’ means with the component. ‘w/o’ means without the component.

performance on most collaborative detectors. We select three typical detectors for comparison here. For instance, our method achieves average improvements of 7.15% in AP@0.3, 7.27% in AP@0.5, and 7.35% in AP@0.7 on V2X-ViT. It also has an improvement of 11.47% in AP@0.3, 11.76% in AP@0.5 and 8.93% in AP@0.7 on F-Cooper, demonstrating the superior performance of the SSDA. When compared with the No Adapt method, the proposed SSDA improves by 16.03% in AP@0.3, 14.5% in AP@0.5 and 12.5% in AP@0.7. It is noteworthy that when using the discriminator as the baseline and combining it with the F-Cooper model, the DUSA only achieves a 1.5% increase in the AP@0.3, whereas the SSDA has an increase of 12.97% in the AP@0.3. The SSDA demonstrates significant performance across different detectors and evaluation metrics, exhibiting strong generalization capabilities.

4.4 Ablation studies

We perform a detailed analysis to explore the effects of various components in the SSDA architecture. The V2X-ViT [48] detector is utilized and the results are performed on the validation set of DAIR-V2X-C.

Method	GAS	LAS	FSA	AP@0.3	AP@0.5	AP@0.7	Mean
Oracle				62.69	59.02	45.61	55.77
No Adapt				34.73	31.23	14.94	26.96
No Adapt w/ GAS	✓			46.85(+12.12)	43.72(+12.49)	28.51(+13.57)	39.69(+12.73)
SSDA w/o LAS		✓		48.57(+13.84)	44.28(+13.05)	25.26(+10.32)	39.37(+12.41)
SSDA w/o FSA			✓	50.91(+16.18)	46.42(+15.19)	24.84(+9.90)	40.55(+13.59)
SSDA	✓	✓	✓	50.76(+16.03)	45.73(+14.50)	27.44(+12.50)	41.43(+14.47)

Table 2: Ablation study of the proposed core components on V2X-ViT. ‘+’ shows the increase in performance compared to the No Adapt method. ‘Mean’ shows the average result of AP@0.3, AP@0.5, and AP@0.7.

Effect of core components. Table 2 summarizes the impact of the core components within the SSDA, i.e., the GAS, the LAS, and the FSA. The consistent improvement in detection proves the effectiveness of each introduced element. Specifically, when using only the GAS, the performance is improved by 12.12% in AP@0.3, 12.49% in AP@0.5 and 13.57% in AP@0.7. Integrating all three components gains greater detection performance by 16.03%, 14.50%, and

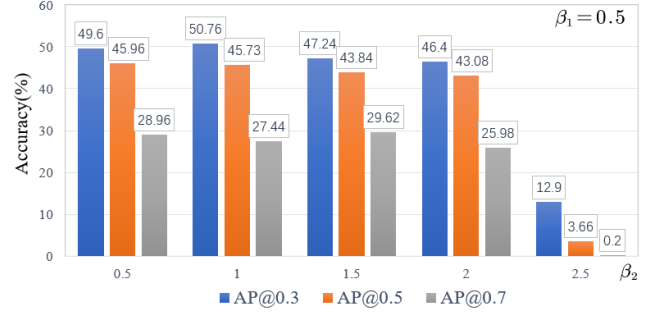


Figure 5: Hyperparameter analysis of β_2 , and the β_1 is 0.5.

12.50% in AP@0.3, AP@0.5 and AP@0.7, respectively. An interesting observation from the results is that when comparing “No Adapt w/ GAS” and “SSDA w/o FSA”, i.e., with the addition of the LAS, the performance improved only at AP@0.3 and AP@0.5, while the performance at AP@0.7 actually decreased. It is mainly attributed to the loss of correlation information between agents. Due to the enforced transfer of local alignment, the semantic information of inter-agent relationships is lost, diminishing the model’s ability to capture vehicle orientation and spatial information. The addition of FSA can mitigate this issue, with an improvement of 2.60% in AP@0.7.

Method	FA	SFW	SAO	AP@0.3	AP@0.5	AP@0.7
SSDA w/o FSA				50.91	46.42	24.84
SSDA w/ SAO			✓	50.15(-0.76)	45.14(- 1.28)	26.57(+1.73)
SSDA w/o FA		✓	✓	49.74(-1.17)	45.51(- 0.91)	26.33(+1.49)
SSDA	✓	✓	✓	50.76(-0.15)	45.73(- 0.69)	27.44(+ 2.60)

Table 3: Ablation study of the main component of FSA on V2X-ViT. FA: Frequency Characteristics Decoupling and Adaptive Feature Refinement. SAO: Statistical Attribute Obfuscation. SFW: Statistical Feature Weighting.

Effect of the components within the FSA: Table 3 summarizes the impact of the core components within the FSA, i.e., the FA, the SFW, and the SAO. The SAO effectively removes the redundant information within the domain and reduces the impact of domain

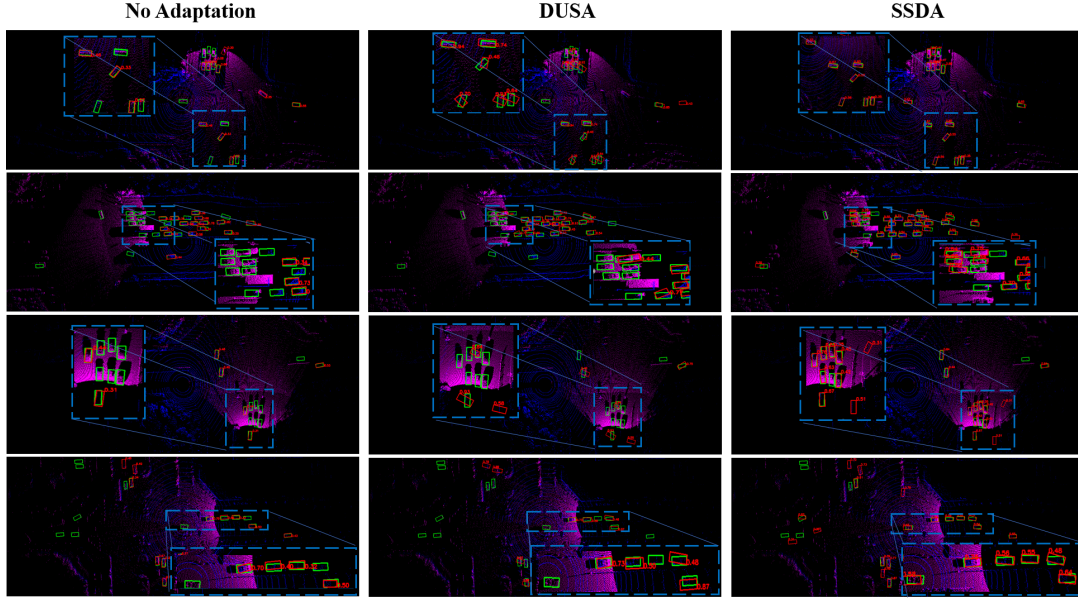


Figure 6: Qualitative comparison results in real-world scenarios from the DAIR-V2X-C dataset. Green and red boxes denote the ground truths and detection results, respectively.

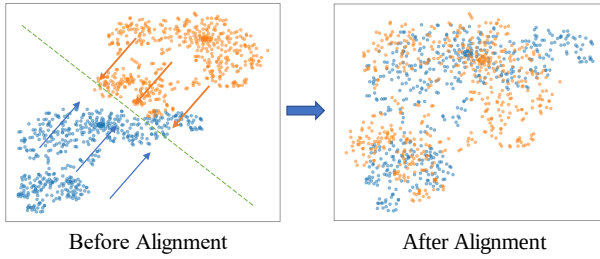


Figure 7: Visualization of latent features with t-SNE [42]. The orange and blue ones indicate the samples from different domains.

interference on spatial localization, thereby ensuring the extraction of spatial position information. As a result, there is a significant improvement in the AP@0.7 metric. Furthermore, the FSA effectively compensated for the absence of spatial position information and semantic information during domain adaptation through frequency information decoupling and weighting. This alleviates the decline in AP@0.3 and AP@0.5 and promotes an increase in AP@0.7.

Hyperparameters settings. Figure 5 explores the impact of the hyperparameters β_1 and β_2 . As β_2 increases, the model pays more attention to local feature alignment and neglects the impact of global alignment. We can see that when β_2 is set to 2.5, the model exhibits poor performance. The primary reason is the relatively reduced proportion of prediction loss and global alignment loss. The SSDA achieves its optimal performance when β_1 is set to 0.5 and β_2 is set to 1. Furthermore, the detailed experimental analysis for hyperparameters β_1 and α are presented in the supplementary materials to ensure the reliability and robustness of the model.

4.5 Qualitative Evaluation

Qualitative Detection Examples. Figure 6 displays the detection results of No adaptation, DUSA [16] and SSDA on DAIR-V2X-C.

The results indicate that the SSDA significantly outperforms the SOTA method DUSA. Many of the missed detections by the DUSA method are successfully detected by our proposed approach. Moreover, SSDA can accurately detect samples even at large distances with sparse point cloud information. These qualitative evaluations confirm the advantages of SSDA in providing accurate and comprehensive perception, especially in scenarios with dense agents.

Latent Feature Visualization. Figure 7 visualizes the distribution changes of feature alignment before and after using t-SNE [42]. The features are effectively aligned into the same data space, and the correlation of the original data does not exhibit significant changes. This demonstrates that the proposed SSDA not only achieves feature alignment effectively but also preserves the underlying associations in the original data.

5 Conclusion

The SSDA method is introduced for multi-agent collaborative perception, containing two key components: the FSA and the SAA. The FSA simplifies feature representation by decoupling correlation features and mitigates the interference for domain adversarial scenarios. The SAA addresses shift uncertainty by achieving adaptive alignment from global to local level, guided by information entropy. By combining SSA with FSA, the SSDA enables adaptive alignment of cross-domain feature distributions while maintaining effective semantic information.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grants 62203040, 62472048, and U2268203, and in part by the Beijing Natural Science Foundation under Grant L242081.

References

- [1] Isabela Albuquerque, João Monteiro, Olivier Rosanne, and Tiago H Falk. 2022. Estimating distribution shifts for predicting cross-subject generalization in electroencephalography-based mental workload assessment. *Frontiers in Artificial Intelligence* 5 (2022), 992732.
- [2] Isabela Albuquerque, João Monteiro, Olivier Rosanne, Abhishek Tiwari, Jean-François Gagnon, and Tiago H Falk. 2019. Cross-subject statistical shift estimation for generalized electroencephalography-based mental workload assessment. In *IEEE International Conference on Systems, Man and Cybernetics*. IEEE, 3647–3653.
- [3] Shai Ben-David, John Blitzer, Koby Crammer, Alex Kulesza, Fernando Pereira, and Jennifer Wortman Vaughan. 2010. A theory of learning from different domains. *Machine Learning* 79 (2010), 151–175.
- [4] Shai Ben-David, John Blitzer, Koby Crammer, and Fernando Pereira. 2006. Analysis of representations for domain adaptation. *Advances in Neural Information Processing Systems* 19 (2006).
- [5] Liling Cao, Xinxin Feng, Junli Liu, and Guofeng Zhou. 2024. Automatic Generation System for Autonomous Driving Simulation Scenarios Based on PreScan. *Applied Sciences* 14, 4 (2024), 1354.
- [6] Gysam Chang, Wonseok Roh, Sujin Jang, Dongwook Lee, Daehyun Ji, Gyeon-grok Oh, Jinsun Park, Jinkyu Kim, and Sangpil Kim. 2024. CMDA: Cross-Modal and Domain Adversarial Adaptation for LiDAR-Based 3D Object Detection. In *AAAI Conference on Artificial Intelligence*, Vol. 38. 972–980.
- [7] Qi Chen, Xu Ma, Sihai Tang, Jingda Guo, Qing Yang, and Song Fu. 2019. F-Cooper: Feature based cooperative perception for autonomous vehicle edge computing system using 3D point clouds. In *ACM/IEEE Symposium on Edge Computing*. 88–100.
- [8] Ching-Yao Chuang, Antonio Torralba, and Stefanie Jegelka. 2020. Estimating Generalization under Distribution Shifts via Domain-Invariant Representations. *Machine Learning Research* 119 (2020).
- [9] Shuhao Cui, Shuhui Wang, Junbao Zhuo, Chi Su, Qingming Huang, and Qi Tian. 2020. Gradually vanishing bridge for adversarial domain adaptation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 12455–12464.
- [10] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. 2017. CARLA: An open urban driving simulator. In *Conference on Robot Learning*. PMLR, 1–16.
- [11] Shahaf E Finder, Yair Zohav, Maor Ashkenazi, and Eran Treister. 2022. Wavelet feature maps compression for image-to-image CNNs. *Advances in Neural Information Processing Systems* 35 (2022), 20592–20606.
- [12] Yaroslav Ganin and Victor Lempitsky. 2015. Unsupervised domain adaptation by backpropagation. In *International Conference on Machine Learning*. PMLR, 1180–1189.
- [13] Shixin Hong, Yu Liu, Zhi Li, Shaohui Li, and You He. 2024. Multi-agent Collaborative Perception via Motion-aware Robust Communication Network. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 15301–15310.
- [14] Yue Hu, Shaoheng Fang, Zixing Lei, Yiqi Zhong, and Siheng Chen. 2022. Where2comm: Communication-efficient collaborative perception via spatial confidence maps. *Advances in Neural Information Processing Systems* 35 (2022), 4874–4886.
- [15] Yue Hu, Juntong Peng, Sifei Liu, Junhao Ge, Si Liu, and Siheng Chen. 2024. Communication-Efficient Collaborative Perception via Information Filling with Codebook. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 15481–15490.
- [16] Xianghao Kong, Wentao Jiang, Jinrang Jia, Yifeng Shi, Runsheng Xu, and Si Liu. 2023. DUSA: Decoupled unsupervised sim2real adaptation for vehicle-to-everything collaborative perception. In *ACM International Conference on Multimedia*. 1943–1954.
- [17] Zhengfeng Lai, Haoping Bai, Haotian Zhang, Xianzhi Du, Jiulong Shan, Yinfei Yang, Chen-Nee Chuah, and Meng Cao. 2024. Empowering unsupervised domain adaptation with large-scale pre-trained vision-language models. In *IEEE/CVF Winter Conference on Applications of Computer Vision*. 2691–2701.
- [18] Jinlong Li, Runsheng Xu, Xinyu Liu, Baolu Li, Qin Zou, Jiaqi Ma, and Hongkai Yu. 2024. S2R-ViT for multi-agent cooperative perception: Bridging the gap from simulation to reality. In *IEEE International Conference on Robotics and Automation*. IEEE, 16374–16380.
- [19] Yiming Li, Dekun Ma, Ziyang An, Zixun Wang, Yiqi Zhong, Siheng Chen, and Chen Feng. 2022. V2X-Sim: Multi-agent collaborative perception dataset and benchmark for autonomous driving. *IEEE Robotics and Automation Letters* 7, 4 (2022), 10914–10921.
- [20] Yiming Li, Shunli Ren, Pengxiang Wu, Siheng Chen, Chen Feng, and Wenjun Zhang. 2021. Learning distilled collaboration graph for multi-agent perception. *Advances in Neural Information Processing Systems* 34 (2021), 29541–29552.
- [21] Zhaowen Li, Xu Zhao, Chaoyang Zhao, Ming Tang, and Jinqiao Wang. 2022. Transferring low-frequency features for domain adaptation. In *2022 IEEE International Conference on Multimedia and Expo (ICME)*. IEEE, 01–06.
- [22] Mingsheng Long, Yue Cao, Jianmin Wang, and Michael Jordan. 2015. Learning transferable features with deep adaptation networks. In *International Conference on Machine Learning*. PMLR, 97–105.
- [23] Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I Jordan. 2018. Conditional adversarial domain adaptation. *Advances in Neural Information Processing Systems* 31 (2018).
- [24] Yifan Lu, Yue Hu, Yiqi Zhong, Dequan Wang, Yanfeng Wang, and Siheng Chen. 2024. An Extensible Framework for Open Heterogeneous Collaborative Perception. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=KkrDUGIAK>
- [25] Guiyang Luo, Chongzhang Shao, Nan Cheng, Haibo Zhou, Hui Zhang, Quan Yuan, and Jinglin Li. 2023. EdgeCooper: Network-aware cooperative LiDAR perception for enhanced vehicular awareness. *IEEE Journal on Selected Areas in Communications* 42, 1 (2023), 207–222.
- [26] Guiyang Luo, Hui Zhang, Quan Yuan, and Jinglin Li. 2022. Complementarity-enhanced and redundancy-minimized collaboration network for multi-agent perception. In *ACM International Conference on Multimedia*. 3578–3586.
- [27] Guiyang Luo, Haibo Zhou, Nan Cheng, Quan Yuan, Jinglin Li, Fangchun Yang, and Xuemin Shen. 2019. Software-defined cooperative data sharing in edge computing assisted 5G-VANET. *IEEE Transactions on Mobile Computing* 20, 3 (2019), 1212–1229.
- [28] Tianyou Luo, Quan Yuan, Guiyang Luo, Yuchen Xia, Yujia Yang, and Jinglin Li. 2025. Plug and Play: A Representation Enhanced Domain Adapter for Collaborative Perception. In *European Conference on Computer Vision*. Springer Nature Switzerland, Cham, 287–303.
- [29] You-Wei Luo and Chuan-Xian Ren. 2023. MOT: Masked Optimal Transport for Partial Domain Adaptation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3531–3540. doi:10.1109/CVPR52729.2023.00344
- [30] You-Wei Luo, Chuan-Xian Ren, Xiao-Lin Xu, and Qingshan Liu. 2024. Geometric Understanding of Discriminability and Transferability for Visual Domain Adaptation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46, 12 (2024), 8727–8742. doi:10.1109/TPAMI.2024.3412680
- [31] Yunsheng Ma, Juanwu Lu, Can Cui, Sicheng Zhao, Xu Cao, Wenqian Ye, and Ziran Wang. 2024. MACP: Efficient model adaptation for cooperative perception. In *IEEE/CVF Winter Conference on Applications of Computer Vision*. 3373–3382.
- [32] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. 2019. Pytorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems* 32 (2019).
- [33] Li Ren, Chen Chen, Liqiang Wang, and Kien Hua. 2024. Towards improved proxy-based deep metric learning via data-augmented domain adaptation. In *AAAI Conference on Artificial Intelligence*, Vol. 38. 14811–14819.
- [34] T-YLPG Ross and GKHP Dollár. 2017. Focal loss for dense object detection. In *IEEE Conference on Computer Vision and Pattern Recognition*. 2980–2988.
- [35] Kuniaki Saito, Shohei Yamamoto, Yoshitaka Ushiku, and Tatsuya Harada. 2018. Open set domain adaptation by backpropagation. In *European Conference on Computer Vision*. 153–168.
- [36] Swami Sankaranarayanan, Yogesh Balaji, Carlos D Castillo, and Rama Chellappa. 2018. Generate to adapt: Aligning domains using generative adversarial networks. In *IEEE Conference on Computer Vision and Pattern Recognition*. 8503–8512.
- [37] Congzhang Shao, Guiyang Luo, Quan Yuan, Yifu Chen, Yilin Liu, Kexin Gong, and Jinglin Li. 2025. Hetecooper: Feature Collaboration Graph for Heterogeneous Collaborative Perception. In *European Conference on Computer Vision*. Springer Nature Switzerland, Cham, 162–178.
- [38] Mostafa Sharifzadeh, Ali K. Z. Tehrani, Habib Benali, and Hassan Rivaz. 2021. Ultrasound Domain Adaptation Using Frequency Domain Analysis. In *2021 IEEE International Ultrasonics Symposium (IUS)*. 1–4.
- [39] Jian Shen, Yanru Qu, Weinan Zhang, and Yong Yu. 2018. Wasserstein distance guided representation learning for domain adaptation. In *AAAI Conference on Artificial Intelligence*, Vol. 32.
- [40] Eric Tzeng, Judy Hoffman, Kate Saenko, and Trevor Darrell. 2017. Adversarial discriminative domain adaptation. In *IEEE Conference on Computer Vision and Pattern Recognition*. 7167–7176.
- [41] Eric Tzeng, Judy Hoffman, Ning Zhang, Kate Saenko, and Trevor Darrell. 2014. Deep domain confusion: Maximizing for domain invariance. *arXiv preprint arXiv:1412.3474* (2014).
- [42] Laurens Van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-SNE. *Journal of Machine Learning Research* 9, 11 (2008).
- [43] Tianhang Wang, Guang Chen, Kai Chen, Zhengfa Liu, Bo Zhang, Alois Knoll, and Changjun Jiang. 2023. UMC: A unified bandwidth-efficient and multi-resolution based collaborative perception framework. In *IEEE/CVF International Conference on Computer Vision*. 8187–8196.
- [44] Yan Wang, Guodong Yin, Peng Hang, Jing Zhao, Yilun Lin, and Chao Huang. 2024. Fundamental estimation for tire road friction coefficient: A model-based learning framework. *IEEE Transactions on Vehicular Technology* (2024).
- [45] Quanmin Wei, Penglin Dai, Wei Li, Bingyi Liu, and Xiao Wu. 2025. CoPEFT: Fast Adaptation Framework for Multi-Agent Collaborative Perception with Parameter-Efficient Fine-Tuning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 23351–23359.
- [46] Wei Wu, Qinwei Fan, Jacek M Zurada, Jian Wang, Dakun Yang, and Yan Liu. 2014. Batch gradient method with smoothing L1/2 regularization for training of

- feedforward neural networks. *Neural Networks* 50 (2014), 72–78.
- [47] Hao Xiang, Runsheng Xu, and Jiaqi Ma. 2023. HM-ViT: Hetero-modal vehicle-to-vehicle cooperative perception with vision transformer. In *IEEE/CVF International Conference on Computer Vision*. 284–295.
 - [48] Runsheng Xu, Hao Xiang, Zhengzhong Tu, Xin Xia, Ming-Hsuan Yang, and Jiaqi Ma. 2022. V2X-ViT: Vehicle-to-everything cooperative perception with vision transformer. In *European Conference on Computer Vision*. Springer, 107–124.
 - [49] Runsheng Xu, Hao Xiang, Xin Xia, Xu Han, Jinlong Li, and Jiaqi Ma. 2022. OPV2V: An open benchmark dataset and fusion pipeline for perception with vehicle-to-vehicle communication. In *International Conference on Robotics and Automation*. IEEE, 2583–2589.
 - [50] Hongliang Yan, Yukang Ding, Peihua Li, Qilong Wang, Yong Xu, and Wangmeng Zuo. 2017. Mind the class weight bias: Weighted maximum mean discrepancy for unsupervised domain adaptation. In *IEEE Conference on Computer Vision and Pattern Recognition*. 2272–2281.
 - [51] Dingkang Yang, Kun Yang, Yuzheng Wang, Jing Liu, Zhi Xu, Rongbin Yin, Peng Zhai, and Lihua Zhang. 2024. How2comm: Communication-efficient and collaboration-pragmatic multi-agent perception. *Advances in Neural Information Processing Systems* 36 (2024).
 - [52] Hongbo Yin, Daxin Tian, Chunmian Lin, Xuting Duan, Jianshan Zhou, Dezong Zhao, and Dongpu Cao. 2025. CUDA-X: Unsupervised Domain-Adaptive Vehicle-to-Everything Collaboration via Knowledge Transfer and Alignment. *IEEE Transactions on Neural Networks and Learning Systems* (2025).
 - [53] Haibao Yu, Yizhen Luo, Mao Shu, Yiyi Huo, Zebang Yang, Yifeng Shi, Zhenglong Guo, Hanyu Li, Xing Hu, Jirui Yuan, and Zaiqing Nie. 2022. Dair-v2x: A large-scale dataset for vehicle-infrastructure cooperative 3d object detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 21361–21370.
 - [54] Jingyu Zhang, Kun Yang, Yilei Wang, Hanqi Wang, Peng Sun, and Liang Song. 2024. ERMVP: Communication-Efficient and Collaboration-Robust Multi-Vehicle Perception in Challenging Environments. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 12575–12584.
 - [55] Siyu Zhang, Lei Su, Jiefei Gu, Ke Li, Weitian Wu, and Michael Pecht. 2024. Category-level selective dual-adversarial network using significance-augmented unsupervised domain adaptation for surface defect detection. *Expert Systems with Applications* 238 (2024), 121879.
 - [56] Yangtao Zheng, Di Huang, Songtao Liu, and Yunhong Wang. 2020. Cross-domain object detection through coarse-to-fine feature adaptation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 13766–13775.